

Q-1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

ANSWER:-

Main Components inside a Desktop PC

- **Motherboard**

The motherboard is the main circuit board that connects and allows communication between all computer components.

- **CPU (Central Processing Unit)**

The CPU is the brain of the computer. It performs calculations and executes program instructions.

- **CPU Cooler (Heatsink and Fan)**

The CPU cooler removes heat from the processor to prevent overheating and maintain stable performance.

- **RAM (Random Access Memory)**

RAM temporarily stores data that the CPU is actively using, allowing programs to run quickly.

- **Storage Drive (HDD/SSD/NVMe SSD)**

The storage drive permanently stores the operating system, software, and user files. SSDs and NVMe SSDs are much faster than HDDs.

- **Power Supply Unit (PSU)**

The PSU converts AC power from the wall into DC power required by the computer's internal components.

- **Graphics Card (GPU)**

The GPU processes graphics, videos, and 3D applications. It may be integrated into the CPU or installed as a dedicated card.

- **Cooling Fans**

Case fans improve airflow inside the cabinet, helping keep the CPU, GPU, and other components cool.

- **Optical Drive (Optional)**

An optical drive reads and writes CDs, DVDs, or Blu-ray discs. Many modern PCs no longer include one.

- **Expansion Cards (Optional)**

Expansion cards, such as Wi-Fi, sound, or network cards, add extra features and functionality to the computer.

- **SATA Data and Power Cables**

These cables connect storage devices to the motherboard and power supply for data transfer and power.

- **CMOS Battery**

The CMOS battery powers the motherboard's clock and preserves BIOS/UEFI settings when the PC is turned off.

- **Cabinet (PC Case)**

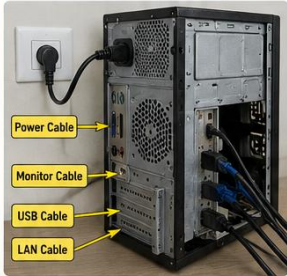
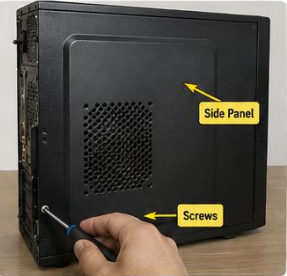
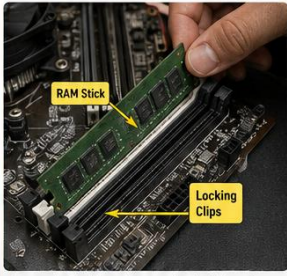
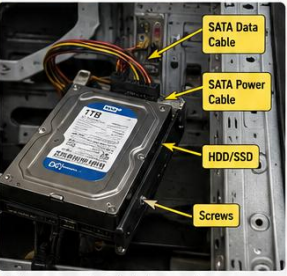
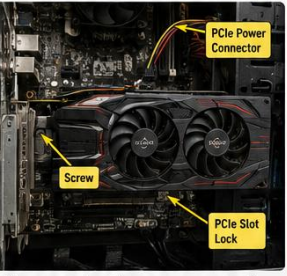
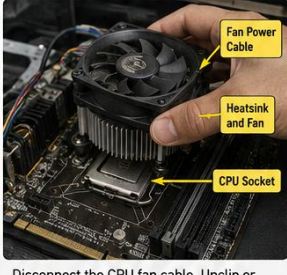
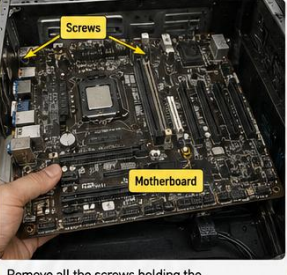
The cabinet houses and protects all internal components while providing mounting points and airflow.

- **Front Panel Connectors**

These connectors link the cabinet's power button, reset button, USB ports, LEDs, and audio ports to the motherboard.

Q-2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.
Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

ANSWER:-

DESKTOP PC DISASSEMBLY STEP-BY-STEP		
1. UNPLUG POWER AND REMOVE ALL CABLES  Turn off the PC. Unplug the power cable from the socket and remove all external cables.	2. REMOVE SIDE PANEL  Remove the screws from the side panel and slide it backward to remove the panel.	3. DISCONNECT INTERNAL CABLES  Disconnect all power cables and data cables connected to the motherboard and drives.
4. REMOVE RAM  Open the locking clips on both sides of the RAM slot. Gently pull the RAM stick up and remove it.	5. REMOVE HDD/SSD  Disconnect the SATA data and power cables. Remove the screws and take out the HDD/SSD from the drive bay.	6. REMOVE GRAPHICS CARD (GPU)  Disconnect the PCIe power connector (if any). Remove the screw at the back. Press the PCIe slot lock and pull the graphics card out.
7. REMOVE CPU COOLER  Disconnect the CPU fan cable. Unclip or unscrew the heatsink and fan assembly and lift it off the CPU.	8. REMOVE MOTHERBOARD  Remove all the screws holding the motherboard. Carefully lift the motherboard up and take it out of the cabinet.	9. ALL COMPONENTS REMOVED  All major components have been removed and are ready for cleaning, upgrading, or replacement.

SAFETY PRECAUTIONS


- Always disconnect power before opening the cabinet.
- Handle components by the edges. Avoid touching connectors.
- Use an anti-static wrist strap if available.
- Keep screws and small parts in a safe place.

Q-3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

ANSWER:-

Step 1: Install the Motherboard

Action:

Place the motherboard inside the cabinet, align it with the standoffs, and secure it using screws.

Explanation:

The motherboard is installed first because it serves as the main platform for connecting all other components.

Step 2: Install the CPU Cooler

Action:

Attach the heatsink and fan to the CPU, then connect the CPU fan cable to the **CPU_FAN** header on the motherboard.

Explanation:

The CPU cooler must be installed before powering on the system to prevent the processor from overheating.

Step 3: Install the RAM Module

Action:

Align the notch on the RAM with the memory slot and press firmly until both locking clips click into place.

Explanation:

RAM is required for the computer to boot and run applications, so it must be installed before testing the system.

Step 4: Install the Storage Drive (HDD/SSD)

Action:

Mount the HDD or SSD in the drive bay, then connect the SATA data cable to the motherboard and the SATA power cable from the power supply.

Explanation:

The storage drive contains the operating system and files, making it essential for booting the computer.

Step 5: Install the Graphics Card (GPU) *(If Available)*

Action:

Insert the graphics card into the PCIe x16 slot, secure it with screws, and connect the PCIe power cable if required.

Explanation:

The GPU provides graphics processing and must be firmly seated to ensure stable video output.

Step 6: Connect Power Supply Cables

Action:

Connect the 24-pin ATX motherboard power cable, 8-pin CPU power cable, SATA power cables, and PCIe power cable (if needed).

Explanation:

Power connections are made after the components are installed to provide electricity safely to every device.

Step 7: Connect Front Panel and Data Cables

Action:

Connect the power switch, reset switch, and power LED, HDD LED, and front USB, front audio, and SATA data cables.

Explanation:

These cables enable the cabinet buttons, USB ports, audio ports, and communication between storage devices and the motherboard.

Step 8: Organize Cables and Close the Cabinet

Action:

Arrange the cables neatly using cable ties, replace the side panel, and tighten the screws.

Explanation:

Good cable management improves airflow, reduces dust buildup, and makes future maintenance easier.

Step 9: Connect External Devices and Power On

Action:

Reconnect the monitor, keyboard, mouse, LAN cable, and power cable. Turn on the PC and verify that it boots successfully.

Explanation:

A successful boot confirms that all components were installed correctly and all connections are secure.

Q-4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

ANSWER:-

1. Safety Precautions Checklist

☐ **Turn off the PC completely**

why: Prevents electrical shock and protects the computer components.

☐ **Unplug the power cable**

why: Ensures there is no electrical power supplied to the system.

☐ **Disconnect all external devices**

why: Prevents cable damage and makes the PC easier to handle.

☐ **Press the power button for 5–10 seconds after unplugging**

why: Discharges any remaining electricity stored in the system.

☐ **Wear an anti-static wrist strap (if available)**

why: Prevents electrostatic discharge (ESD), which can damage sensitive electronic components.

☐ **Work on a clean, dry, and non-conductive surface**

why: Reduces the risk of electrical damage and keeps components clean.

☐ **Handle components by their edges**

why: Avoids damaging chips, circuits, and gold connectors.

☐ **Keep screws in a small container**

why: Prevents screws and small parts from getting lost.

☐ **Avoid touching motherboard circuits and gold connectors**

why: Protects delicate electronic parts from static electricity and physical damage.

☐ **Keep food and liquids away from the workspace**

why: Prevents accidental spills that could damage the computer.

2. Tools Needed Checklist

☐ **Phillips-head screwdriver**

why: Used to remove and tighten most screws inside the PC.

☐ **Anti-static wrist strap**

why: Protects internal components from electrostatic discharge (ESD).

☐ **Plastic pry tool (optional)**

why: Helps remove panels or connectors safely without scratching them.

☐ **Small container or magnetic tray**

why: Keeps screws and small parts organized during disassembly.

☐ **Flashlight or LED light**

why: Makes it easier to see small connectors and components inside the cabinet.

☐ **Cable ties (Zip ties)**

why: Keeps cables neat and improves airflow after reassembly.

☐ **Soft microfiber cloth**

why: Used to clean dust and fingerprints from components.

☐ **Compressed air can (optional)**

why: Safely removes dust from fans, heatsinks, and other internal parts.

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